

DIABETES PREDICTION USING MACHINE LEARNING MODELS KNN & LOGISTIC REGRESSION

Data Science PITP Phase-II Batch-V (Group:1) Weekend Program

ABSTRACT: This project aims to predict diabetes using machine learning techniques based on medical and demographic data. K-Nearest Neighbors (KNN) and Logistic Regression models were applied after dimensionality reduction using PCA. The models were evaluated using multiple performance metrics, with special focus on handling class imbalance. The study highlights strengths and weaknesses of both models and provides insights for improvement.

DATASET DESCRIPTION:

The dataset was collected from Kaggle and contains medical and demographic information of patients, including age, gender, BMI, hypertension, heart disease, smoking history, HbA1c level, and blood glucose level. The target variable indicates whether a patient has diabetes (1) or not (0). This dataset is slightly imbalanced, with fewer diabetic cases compared to non-diabetic cases.

METHODOLOGY:

Data preprocessing steps included handling missing values, encoding categorical variables, and feature scaling. Principal Component Analysis (PCA) was applied to reduce dimensionality and improve computational efficiency. Two classification models were implemented:

1. K-Nearest Neighbors (KNN)
2. Logistic Regression

Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

RESULTS AND FINDINGS:

Both models achieved high overall accuracy (above 96%), primarily due to the dominance of the nondiabetic class. However, deeper analysis shows significant differences in handling the minority class (diabetic patients).

KNN outperformed Logistic Regression in detecting diabetic cases. It achieved higher recall (0.43 vs 0.35) and F1score (0.59 vs 0.49), indicating better balance between precision and recall. Logistic Regression showed lower sensitivity, missing more actual diabetic cases (higher false negatives).

This suggests that while Logistic Regression is efficient and interpretable, KNN is more effective in identifying positive cases in this dataset.

INSIGHTS:

- Recall and F1-score are more important for medical predictions.
- KNN captures complex patterns better than Logistic Regression in this case.
- PCA helped reduce dimensionality without significantly affecting performance.

LIMITATIONS:

- The dataset is imbalanced, which affects model performance on minority class.
- KNN is computationally expensive for large datasets.
- Logistic Regression assumes linear relationships, which may not capture complex patterns.
- PCA reduces interpretability of features.
- No hyperparameter tuning was extensively performed.

FUTURE IMPROVEMENTS:

- Apply techniques like SMOTE to handle class imbalance.
- Perform hyperparameter tuning (GridSearchCV).
- Try advanced models like Random Forest, XGBoost, or Neural Networks.
- Use feature selection instead of only PCA.
- Collect more balanced and diverse data.

CONCLUSION:

This project demonstrates the application of machine learning in healthcare prediction tasks. Although both models performed well, KNN proved to be more effective in identifying diabetic patients. However, further improvements are needed to enhance model reliability, especially for real-world clinical use.

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Github Colaboratory:

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